

Zhidan Liu

Assistant Professor
Thrust of Intelligent Transportation, System Hub
Thrust of Artificial Intelligence, Information Hub
The Hong Kong University of Science and Technology (Guangzhou), China

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EMPLOYMENT

- 2024.08 - Now Assistant Professor, **The Hong Kong University of Science and Technology (Guangzhou)**, China
- 2022.07 - 2024.07 Associate Professor, **Shenzhen University**, China
- 2017.07 - 2022.06 Assistant Professor, **Shenzhen University**, China
- 2015.01 - 2017.06 Research Fellow, **Nanyang Technological University**, Singapore (Advisor: *Prof. Mo Li*)

EDUCATION

- 2009.09 - 2014.09 Ph.D. in Computer Science, **Zhejiang University**, Hangzhou, China, Advisor: *Prof. Dongming Lu*
- 2012.10 - 2013.11 Visiting student, **Nanyang Technological University**, Singapore, Advisor: *Prof. Mo Li*
- 2005.09 - 2009.07 B.E. in Computer Science, **Northeastern University**, Shenyang, China

RESEARCH INTERESTS

Artificial Internet of Things, Urban Computing, Mobile/Edge Intelligence, Data Mining and Analytic

HONORS AND AWARDS

- 2025.08 CCF Transactions on Pervasive Computing and Interaction *Best Reviewer Award*
- 2021.04 Outstanding Undergraduate Teacher Award (Shenzhen University)
- 2020.12 Excellent Class Director of Shenzhen University Tencent Mentor Award
- 2020.12 IEEE ICPADS'20 Best Paper Award
- 2018.12 Award of Tencent "Rhinoceros Birds" - Scientific Research Foundation for Young Teachers
- 2018.06 Shenzhen Nanshan District Pilot Talent Award
- 2018.01 Shenzhen Peacock Plan Talent Award

PUBLICATIONS

([†] indicates the students I advised or co-advised, * indicates the corresponding author)

Refereed Journal Publications

1. [TCSS] Haodong Liu, Ye Zhang, **Zhidan Liu**, Hongyuan Zhu, Changhong Wang. "Heterogeneous Multi-Modal Multi-Label Behavioral Context Recognition Using Wearable Sensors and Mobile Devices", *IEEE Transactions on Computational Social Systems*, Vol. xx, Issue xx, Pages xxxx-xxxx, March 2026. (IF: 4.900, JCR Q1)
2. [TITS] Jiyu Wang[†], **Zhidan Liu***. "A Two-Stage Imputation Method for Urban Traffic Data Based on Sparse Spatiotemporal Attention", *IEEE Transactions on Intelligent Transportation Systems*, Vol. 27, Issue 8, Pages 9329-9344, August 2026. (IF: 8.400, [CCF-Rank B](#), JCR Q1)
3. [TMC] Zhenyu Yang[†], Qiang Yang, **Zhidan Liu***, Zhenjiang Li, Yongpan Zou, Kaishun Wu. "ArmPad: Transforming Forearms into Interaction Interfaces with Smartwatches", *IEEE Transactions on Mobile Computing*, Vol. 25, Issue 8, Pages 12750-12763, August 2026. (IF: 9.200, [CCF-Rank A](#), JCR Q1)
4. [TMC] Xiaosi Liu[†], Xiaowen Xu[†], **Zhidan Liu***, Zhenjiang Li, Kaishun Wu. "Spatio-Temporal Diffusion Model for Cellular Traffic Generation", *IEEE Transactions on Mobile Computing*, Vol. 25, Issue 1, Pages 257-271, January 2026. (IF: 9.200, [CCF-Rank A](#), JCR Q1)
5. [TITS] **Zhidan Liu**, Zhengze Sun[†], Junru Zhang, Bolin Zhang[†], Panrong Tong. "Regional Knowledge Transfer for Urban Traffic Flow Prediction via Satellite Imagery Assisted Contrastive Domain Adaptation", *IEEE Transactions on Intelligent Transportation Systems*, Vol. 26, Issue 11, Pages 19120-19134, November 2025. (IF: 8.400, [CCF-Rank B](#), JCR Q1)
6. [TMC] Zixiao Wang[†], Qi Dong, Tianzhang Xing*, **Zhidan Liu***, Zhenjiang Li, Xiaojiang Chen. "Enabling Effective OOD Detection via Plug-and-Play Network for Mobile Visual Applications", *IEEE Transactions on Mobile Computing*, Vol. 24, Issue 11, Pages 12471-12486, November 2025. (IF: 7.700, [CCF-Rank A](#), JCR Q1)
7. [TMC] Junru Zhang, Cheng Peng, **Zhidan Liu**, Lang Feng, Yuhang Wu, Yabo Dong, Duanqing Xu. "DI2SDiff++: Activity Style Decomposition and Diffusion-Based Fusion for Cross-Person Generalization in Activity Recognition", *IEEE Transactions on Mobile Computing*, Vol. 24, Issue 10, Pages 10760-10777, October 2025. (IF: 7.700, [CCF-Rank A](#), JCR Q1)

8. [TMC] **Zhidan Liu**, Guofeng Ouyang[†], Bolin Zhang[†], Bo Du, Chao Chen, Kaishun Wu. “Joint Order Dispatching and Vehicle Repositioning for Dynamic Ridesharing”, *IEEE Transactions on Mobile Computing*, Vol. 24, Issue 4, Pages 2628-2643, April 2025. (IF: 7.700, **CCF-Rank A**, JCR Q1)
9. [TKDE] **Zhidan Liu**, Yingqian Zhou[†], Xiaosi Liu[†], Haodi Zhang, Yabo Dong, Dongming Lu, Kaishun Wu. “Learning Road Network Index Structure for Efficient Map Matching”, *IEEE Transactions on Knowledge and Data Engineering*, Vol. 37, Issue 1, Pages 423-437, January 2025. (IF: 8.900, **CCF-Rank A**, JCR Q1)
10. [TBD] Biyao Gong[†], Tianzhang Xing*, **Zhidan Liu***, Wei Xi, Xiaojiang Chen. “Adaptive Client Clustering for Efficient Federated Learning over Non-IID and Imbalanced Data”, *IEEE Transactions on Big Data*, Vol. 10, Issue 6, Pages 1051-1065, December 2024. (IF: 3.344, JCR Q1)
11. [TMC] Xianjing Wu[†], **Zhidan Liu***, Zhenjiang Li, Shengjie Zhao*. “Greta: Towards A General Roadside Unit Deployment Framework”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 7, Pages 7602-7617, July 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
12. [TMC] Biyao Gong[†], Tianzhang Xing*, **Zhidan Liu***, Wei Xi, Xiaojiang Chen. “Towards Hierarchical Clustered Federated Learning with Model Stability on Mobile Devices”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 6, Pages 7148-7164, June 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
13. [TMC] **Zhidan Liu**, Hongquan Zhang[†], Guofeng Ouyang[†], Junyang Chen, Kaishun Wu. “Data-Driven Pick-Up Location Recommendation for Ride-Hailing Services”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 2, Pages 1001-1015, February 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
14. [TMC] **Zhidan Liu**, Jiancong Liu[†], Xiaowen Xu[†], Kaishun Wu. “DeepGPS: Deep Learning Enhanced GPS Positioning in Urban Canyons”, *IEEE Transactions on Mobile Computing*, Vol. 23, Issue 1, Pages 376-392, January 2024. (IF: 7.900, **CCF-Rank A**, JCR Q1)
15. [TMC] **Zhidan Liu**, Junhong Zheng[†], Jinye Lin[†], Liang Wang, Kaishun Wu. “Radar: Adversarial Driving Style Representation Learning with Data Augmentation”, *IEEE Transactions on Mobile Computing*, Vol. 22, No. 12, Pages 7070-7085, December 2023. (IF: 7.900, **CCF-Rank A**, JCR Q1)
16. [MONE] Biyao Gong[†], Tianzhang Xing*, **Zhidan Liu***, Junfeng Wang, Xiuya Liu. “Adaptive Clustered Federated Learning for Heterogeneous Data in Edge Computing”, *Mobile Networks and Applications*, DOI: <https://doi.org/10.1007/s11036-022-01978-8>, 2022. (IF: 3.426, JCR Q2)
17. [CJC] **Zhidan Liu**, Weixin Lin[†], Kaishun Wu. “A Spatial Structure Matching Algorithm for Large Spatial-Textual Datasets” (in Chinese), *Chinese Journal of Computers*, Vol. 45, Issue 6, Pages 1261-1275, June 2022. (**CCF-Rank A**)
18. [TIST] Liang Wang, Zhiwen Yu, Bin Guo, Dingqi Yang, Lianbo Ma, **Zhidan Liu**, Fei Xiong. “Data-Driven Targeted Advertising Recommendation System for Outdoor Billboard”, *ACM Transactions on Intelligent Systems and Technology*, Vol. 13, No. 2, Article 29, Pages 1-23, January 2022. (IF: 10.489, JCR Q1)
19. [IOTJ] **Zhidan Liu**, Zengyang Gong[†], Jiangzhou Li[†], Kaishun Wu, “mT-Share: A Mobility-Aware Dynamic Taxi Ridesharing System”, *IEEE Internet of Things Journal*, Vol. 9, Issue 1, Pages 182-198, January 2022. (IF: 10.238, JCR Q1)
20. [TITS] **Zhidan Liu**, Jiangzhou Li[†], Kaishun Wu. “Context-Aware Taxi Dispatching at City-Scale Using Deep Reinforcement Learning”, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 23, Issue 3, Pages 1996-2009, March 2022. (IF: 9.551, **ESI Highly Cited Paper**, **CCF-Rank B**, JCR Q1)
21. [IOTJ] Shiwei Song, **Zhidan Liu**, Zhenjiang Li, Tianzhang Xing, Dingyi Fang. “Coverage-Oriented Task Assignment for Mobile Crowdsensing”, *IEEE Internet of Things Journal*, Vol. 7, Issue 8, Pages 7407-7418, April 2020. (IF: 9.936, JCR Q1)
22. [IOTJ] **Zhidan Liu**, Zhenjiang Li, Kaishun Wu. “UniTask: A Unified Task Assignment Design for Mobile Crowdsourcing based Urban Sensing”, *IEEE Internet of Things Journal*, Vol.6 Issue 4, Pages 6629-6641, August 2019. (IF: 9.936, JCR Q1)
23. [TMC] **Zhidan Liu**, Pengfei Zhou, Zhenjiang Li, Mo Li. “Think Like A Graph: Real-Time Traffic Estimation at City-Scale”, *IEEE Transactions on Mobile Computing*, Vol. 18, Issue 10, Pages 2446-2459, October 2019. (IF: 5.112, **CCF-Rank A**, JCR Q1)
24. [IEEENet] **Zhidan Liu**, Zhenjiang Li, Kaishun Wu, Mo Li. “Urban Traffic Prediction from Mobility Data Using Deep Learning”, *IEEE Network Magazine*, Vol. 32, Issue 4, Pages 40-46, August 2018. (IF: 8.808, JCR Q1)

25. [TITS] **Zhidan Liu**, Shiqi Jiang, Pengfei Zhou, and Mo Li. “A Participatory Urban Traffic Monitoring System: The Power of Bus Riders”, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 18, Issue 10, Pages 2851-2864, October 2017. (IF: 4.051, [CCF-Rank B](#), JCR Q1)
26. [TON] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, and Dongming Lu. “Path Reconstruction in Dynamic Wireless Sensor Networks Using Compressive Sensing”, *IEEE/ACM Transactions on Networking*, Vol. 24, Issue 4, Pages 1948-1960, August 2016. (IF: 3.376, [CCF-Rank A](#), JCR Q1)
27. [TITS] **Zhidan Liu**, Zhenjiang Li, Mo Li, Wei Xing, and Dongming Lu. “Mining Road Network Correlation for Traffic Estimation via Compressive Sensing”, *IEEE Transactions on Intelligent Transportation Systems*, Vol. 17, Issue 7, Pages 1880-1893, July 2016. (IF: 3.724, [CCF-Rank B](#), JCR Q1)
28. [IJDSN] Bo Zeng, Yabo Dong, **Zhidan Liu**. “A Negotiation-Based TDMA Scheduling with Consecutive Slots Assignment for Wireless Sensor Networks”, *International Journal of Distributed Sensor Networks*, 2014. (IF: 1.239, JCR Q4)
29. [IJDSN] **Zhidan Liu**, Wei Xing, Yongchao Wang, Dongming Lu. “Hierarchical Spatial Clustering in Multihop Wireless Sensor Networks”, *International Journal of Distributed Sensor Networks*, 2013. (IF: 1.239, JCR Q4)

Refereed Conference Publications

1. [CIKM’26] Guangyu Wang[†], **Zhidan Liu***. “RiskTraf: Risk-Extrapolated Residual Learning for Multi-Variate Traffic Flow Prediction”, In *ACM CIKM*, Rome, Italy, November 2026. (Acceptance Rate: $597/2216 = 27\%$, [CCF-Rank B](#))
2. [MobiCom’26] Jun You, Kun Wang, Jiesong Chen, **Zhidan Liu**, Zhenjiang Li. “ExoMem: OS-Governed Memory for Local Large-LLM Inference on Mobile Edge Devices”, In *ACM MobiCom*, Austin Texas, USA, October 2026. ([CCF-Rank A](#), [CSRanking Conference](#))
3. [KDD’26] Fansheng Sun[†], Jiyu Wang[†], **Zhidan Liu***. “Orchestrating Reasoning and Reaction: An Asynchronous Hierarchical Framework for LLM-driven Traffic Signal Control”, In *ACM KDD*, Jeju, South Korea, August 2026. (Acceptance Rate: $\sim 18.5\%$, [CCF-Rank A](#), [CSRanking Conference](#))
4. [ACL’26] Quyu Kong, Xu Zhang, Zhenyu Yang[†], Nolan Gao, Chen Liu, Panrong Tong, chenglin cai, Hanzhang Zhou, Jianan Zhang, Liangyu Chen, **Zhidan Liu**, Steven Hoi, Yue Wang. “MobileWorld: Benchmarking Autonomous Mobile Agents in Agent-User Interactive and MCP-Augmented Environments”, In *ACL (Main Conference)*, San Diego, California, United States, July 2026. (Acceptance Rate: 19% , [CCF-Rank A](#), [CSRanking Conference](#))
5. [ACL’26] Haodi Zhang, Xinrui Zhu, Mingze Kong, **Zhidan Liu**, Tao Fan, Kaishun Wu, Yuanfeng Song. “MTO: A Multi-turn Conversational Text-to-OverpassQL Dataset for Enhanced OpenStreetMap Query Generation”, In *ACL (Findings)*, San Diego, California, United States, July 2026. ([CSRanking Conference](#))
6. [GPC’25] Yaqi Huang[†], Zhengze Sun[†], **Zhidan Liu***. “Accelerating Graph Similarity Search with Maximum Common Subgraphs”, In *The 20th International Conference on Green, Pervasive, and Cloud Computing (GPC)*, Kuala Lumpur, Malaysia, December 2025.
7. [CIKM’25] Bolin Zhang[†], **Zhidan Liu***. “DSETA: Driving Style-Aware Estimated Time of Arrival”, In *ACM CIKM*, Seoul, Korea, November 2025. (Acceptance Rate: $443/1627 = 27\%$, [CCF-Rank B](#))
8. [ICDM’24] **Zhidan Liu**, Jinye Lin[†], Zhiyu Xia[†], Chao Chen, Kaishun Wu. “Towards Efficient Ridesharing via Order-Vehicle Pre-Matching Using Attention Mechanism”, In *IEEE ICDM*, Abu Dhabi, UAE, December 2024. (Acceptance Rate: $66/604 = 11\%$, [CCF-Rank B](#))
9. [KDD’24] Junru Zhang, Lang Feng, **Zhidan Liu**, Yuhan Wu, Yang He, Yabo Dong, Duanqing Xu. “Diverse Intra- and Inter-Domain Activity Style Fusion for Cross-Person Generalization in Activity Recognition”, In *ACM KDD*, Barcelona, Spain, August 2024. (Acceptance Rate: 20% , [CCF-Rank A](#))
10. [ICPADS’23] Meiquan Lai[†], Yaqi Huang[†], **Zhidan Liu***, Kaishun Wu. “An Optimized Lossless Graph Summarization for Large-Scale Graphs”, In *IEEE ICPADS*, Ocean Flower Island, Hainan, China, December 2023.
11. [MM’23] Mengzhu Wang, Junyang Chen, Huan Wang, Huisi Wu, **Zhidan Liu**, Qin Zhang. “Interpolation Normalization for Contrast Domain Generalization”, In *ACM MM*, Ottawa, Canada, October 2023. ([CCF-Rank A](#))
12. [ICPADS’22] Jianlong Xu, Zhiyu Xia[†], Yuhui Li, Yuxiang Zeng, **Zhidan Liu**. “Subgraph Sampling for Inductive Sparse Cloud Services QoS Prediction”, In *IEEE ICPADS*, Nanjing, China, January 2023.

13. [DASFAA'21] Zhidan Liu, Junhong Zheng[†], Zengyang Gong[†], Haodi Zhang, Kaishun Wu. "Exploiting Multi-Source Data for Adversarial Driving Style Representation Learning", In *DASFAA*, Taipei, Taiwan, April 2021. (CCF-Rank B)
14. [HPCC'20] Baofu Huang[†], Zhidan Liu*, Kaishun Wu. "Structure Preserved Graph Reordering for Fast Graph Processing Without the Pain", In *IEEE HPCC*, Fiji, December 2020. (Acceptance Rate: 58/379 = 15%)
15. [ICPADS'20] Baofu Huang[†], Zhidan Liu*, Kaishun Wu. "Accelerating PageRank in Shared-Memory for Efficient Social Network Graph Analytics", In *IEEE ICPADS*, HongKong, China, December 2020. (Best Paper Award)
16. [ICCCN'20] Zengyang Gong[†], Bo Du, Zhidan Liu, Wei Zeng, Pascal Perez, Kaishun Wu. "SD-seq2seq: A Deep Learning Model for Bus Bunching Prediction Based on Smart Card Data", In *IEEE ICCCN*, Honolulu, Hawaii, USA, August 2020.
17. [ICDE'20] Zhidan Liu, Zengyang Gong[†], Jiangzhou Li[†], Kaishun Wu. "Mobility-Aware Dynamic Taxi Ridesharing", In *IEEE ICDE*, Dallas, Texas, USA, April 2020. (CCF-Rank A, CSRanking Conference)
18. [MSN'19] Zhidan Liu, Mingliang Huang[†], Zhi Ye[†], Kaishun Wu. "DeepRTP: A Deep Spatio-Temporal Residual Network for Regional Traffic Prediction", In *IEEE MSN*, Hong Kong, China, December 2019. (Invited paper)
19. [MobiSys'19] Yang Liu, Zhenjiang Li, Zhidan Liu, Kaishun Wu. "Real-time Arm Skeleton Tracking and Gesture Inference Tolerant to Missing Wearable Sensors", In *ACM MobiSys*, Seoul, Republic of Korea, June 2019. (Acceptance Rate: 40/174 = 22%, CCF-Rank B, CSRanking Conference)
20. [IPSN'18] Chu Cao, Zhidan Liu, Mo Li, Qin Zheng, Wenqiang Wang. "Walkway Discovery from Large Scale Crowdsensing", In *ACM/IEEE IPSN*, Porto, Portugal, April 2018. (Acceptance Rate: 22/82 = 26%, CCF-Rank B)
21. [MobiHoc'14] Zhidan Liu, Zhenjiang Li, Mo Li, Wei Xing, Dongming Lu. "Path Reconstruction in Dynamic Wireless Sensor Networks using Compressive Sensing", In *ACM MobiHoc*, Philadelphia, USA, August 2014. (Acceptance Rate: 40/211 = 18%, CCF-Rank B)
22. [ICCC'13] Zhidan Liu, Wei Xing, Yongchao Wang, Dongming Lu. "Hierarchical Spatial Clustering in Multi-hop Wireless Sensor Networks", In *IEEE ICC*, Xi'an, China, August 2013.
23. [AINA'13] Zhidan Liu, Wei Xing, Bo Zeng, Yongchao Wang, Dongming Lu. "Distributed Spatial Correlation-based Clustering for Approximate Data Collection in WSNs", In *IEEE AINA*, Barcelona, Spain, March 2013.
24. [ICACT'13] Zhidan Liu, Wei Xing, Yongchao Wang, Dongming Lu. "An Energy-Efficient Data Collection Scheme for Wireless Sensor Networks", In *IEEE ICACT*, PyeongChang, Korea, January 2013.
25. [CyberC'12] Bo Zeng, Yabo Dong, Zhidan Liu, Dongming Lu. "A Workload-Aware Link Scheduling for Heterogeneous Wireless Sensor Networks", In *IEEE CyberC*, Sanya, China, October 2012.
26. [WiCom'12] Zhidan Liu, Wei Xing, Yongchao Wang, Dongming Lu. "Adaptive Data Storage for Querying in Wireless Sensor Networks", In *IEEE WiCom*, Shanghai, China, September 2012.

Books, Poster, and Demo

1. [MobiCom'25] Xiaowen Xu, Xiaosi Liu, Zhidan Liu*, Zhenjiang Li, Kaishun Wu. "Poster: Diffusion-Driven Spatio-Temporal Modeling of Cellular Traffic Generation", In *ACM MobiCom*, Hong Kong, China, November 2025.
2. [Book'21] Zhidan Liu, Kaishun Wu. "Mobility Data-Driven Urban Traffic Monitoring", *Springer*, Singapore, May, 2021.
3. [MobiSys'19] Yang Liu, Chengdong Lin, Zhenjiang Li, Zhidan Liu, Kaishun Wu. "Poster: When Wearable Sensing Meets Arm Tracking", In *ACM MobiSys*, Seoul, Republic of Korea, June 2019.
4. [IPSN'18] Chu Cao, Zhidan Liu, Mo Li, Qin Zheng, Wenqiang Wang. "Demo Abstract: Walkway Discovery from Large Scale Crowdsensing", In *ACM/IEEE IPSN*, Porto, Portugal, April 2018.

Patents

1. Zhidan Liu, Meiquan Lai, Kaishun Wu. *Lossless Graph Data Compression Method Based on Improved Graph Summarization Algorithm* (ZL202310062494.5), 2026.06.17, Chinese invention patent
2. Zhidan Liu, Yingqian Zhou, Kaishun Wu. *A Method for Adaptively Adjusting the Search Radius in Map Matching Based on Reinforcement Learning* (ZL202310321723.0), 2026.02.06, Chinese invention patent

3. **Zhidan Liu**, Jiayue Ruan, Kaishun Wu. *Method and System for Adversarial Domain Adaptation-Based Human Activity Recognition Using IMU Data* (ZL202310336403.2), 2025.10.31, Chinese invention patent
4. **Zhidan Liu**, Yingqian Zhou, Kaishun Wu. *A Method for Constructing Road Network Spatial Index Based on Learned Index* (ZL202310321722.6), 2025.10.03, Chinese invention patent
5. **Zhidan Liu**, Yaqi Huang. *Graph Similarity Search Method, Device, Computer-Readable Storage Medium, and Program* (ZL202410540182.5), 2025.09.12, Chinese invention patent
6. **Zhidan Liu**, Weixin Lin, Kaishun Wu. *A Spatial Structure Matching Method for Large Spatial-Textual Datasets* (ZL202111190044.1), 2024.12.31, Chinese invention patent
7. **Zhidan Liu**, Jiancong Liu, Kaishun Wu. *A Deep Learning-Based GPS Enhancement Positioning Method and System for Urban Canyons* (ZL202210425764.X), 2024.10.29, Chinese invention patent
8. **Zhidan Liu**, Baofu Huang, Kaishun Wu. *A Lightweight and Efficient Method for Vertex Reordering in Graphs* (ZL202011134445.0), 2023.12.12, Chinese invention patent
9. **Zhidan Liu**, Baofu Huang, Mingliang Huang, Kaishun Wu. *Parallel Computing Acceleration Method for PageRank on Social Network Graph Data* (ZL201910999376.0), 2023.10.13, Chinese invention patent
10. **Zhidan Liu**, Junhong Zheng, Kaishun Wu. *A Method for Learning Driving Style Representation based on Multi-context Data* (ZL202011431508.9), 2023.01.13, Chinese invention patent
11. **Zhidan Liu**, Kaishun Wu. *A Unified Mobile Crowdsourcing Task Allocation Method for Urban Sensing* (ZL201910121165.7), 2022.09.20, Chinese invention patent
12. **Zhidan Liu**, Zenyang Gong, Jiangzhou Li, Kaishun Wu. *A Mobility-Aware Dynamic Ridesharing System* (ZL201910515242.7), 2022.09.20, Chinese invention patent
13. **Zhidan Liu**, Pengfei Zhou, Zhenjaing Li, Mo Li, Kaishun Wu. *A Traffic Estimation Method for Urban Road Networks* (ZL201811140012.9), 2021.09.24, Chinese invention patent
14. **Zhidan Liu**, Jiangzhou Li, Kaishun Wu. *A Deep Reinforcement Learning based Taxi Dispatching Method* (ZL202010521882.1), 2021.07.13, Chinese invention patent

SELECTED PROJECTS/GRANTS

- [PI] **Dynamic Optimization and Cooperative Decision-Making Mechanisms for Crowd-Powered Shared Mobility**, **National Natural Science Foundation of China (General Program)**, No.62572416, 2026.01 - 2029.12, 500K CNY. (*Ongoing*)
- [PI] **Research on the Key Technologies of Mobility Context Aware Personalized Ridesharing**, **National Natural Science Foundation of China (General Program)**, No.62172284, 2022.01 - 2025.12, 752K CNY. (Completed)
- [PI] **Research on Exploiting Big Mobility Trajectory Data for Smart Sensing of Urban Traffic**, **National Natural Science Foundation of China (Young Scientists Fund)**, No.61802261, 2019.01 - 2021.12, 300K CNY. (Completed)
- [PI] **Research on the Key Technologies of Shared Memory based Social Network Graph Computing**, **National Natural Science Foundation of Guangdong (General Program)**, No.2022A1515010155, 2022.01 - 2024.12, 100K CNY. (Completed)
- [PI] **Joint Research of Crowdsensing and Data Analytic for Smart City**, **National Natural Science Foundation of Guangdong (General Program)**, No.2020A1515011502, 2019.10 - 2022.09, 100K CNY. (Completed)
- [PI] **Research on Crowdsensing based Urban Sensing Framework**, **Natural Science Foundation of Shenzhen University Grant**, No.2018061, 2018.01 - 2022.12, 200K CNY. (Completed)
- [PI] **Research on Big Mobility Trajectory Data based Intelligent Transport Systems**, **Startup Fund for Shenzhen Overseas High-Caliber Personnel**, No.827/000344, 2018.01 - 2021.12, 2700K CNY. (Completed)
- [PI] **Research on Exploiting Big Trajectory Data for Intelligent Transport Sensing and Computing**, **Tencent "Rhinoceros Birds" - Scientific Research Foundation for Young Teachers of Shenzhen University**, No.201802, 2019.01 - 2020.12, 50K CNY. (Completed)

- [Co-I] **Key Innovation Team on Meteorological Service Support Technologies for Low-Altitude Flight, Science and Technology Innovation Team of Guangdong Provincial Meteorological Bureau**, GRMCTD02505-ZD05, 2025.09 - 2028.09, 67.5K CNY. (*Ongoing*)
- [Co-I] **Multi-dimensional and Deep Perception Information based Social Computing, Joint Funds of National Natural Science Foundation of China (Key Program)**, No.U2001207, 2021.01 - 2024.12, 2550K CNY. (Completed)

INVITED TALKS

- “*AI-Powered Intelligent Transportation: From Urban Perception to Smart Decisions*”, CAAI BDSC 2026, Chongqing, China, July 2026
- “*AI-Driven Dynamic Ridesharing*”, HMCC 2025 & HHME 2025, Dalian, China, August 2025
- “*AI-Driven Dynamic Ridesharing*”, UGOD 2025 Summer AI Symposium, HKUST (GZ), Guangzhou, China, July 2025
- “*The New Era of Artificial Intelligence: Opportunities and Challenges*”, Guangzhou Foreign Language School, Guangzhou, February 2025
- “*Data-Driven Pick-Up Location Recommendation for Ride-Hailing Services*”, CCF HHME 2024, Taiyuan, China, August 2024
- “*Deep Learning Enhanced GPS Positioning in Urban Canyons*”, Embedded AI Summit 2024, Dunhuang, China, August 2024
- “*Trajectory Data Driven Smart Mobility: Sensing, Computing, and Applications*”, HKUST (GZ), Guangzhou, China, September 2023
- “*Trajectory Data Driven Smart Transportation Research: Sensing and Computing*”, Zhejiang University, Hangzhou, China, April 2023
- “*Trajectory Data Driven Smart Transportation Research: Sensing and Computing*”, online, CCF HHME 2022, September 2022

TEACHING

Teaching Courses (at HKUST (Guangzhou)):

- | | |
|--|------------------|
| • (UG) C++ Programming (UFUG 2601) | Fall 2025/2026 |
| • (UG) Introduction to Data Mining (AIAA 3111) | Fall 2026 |
| • (PG) Mobile Sensing and Computing for Transportation (INTR 6000N) | Spring 2025/2026 |

Teaching Courses (at Shenzhen University):

- | | |
|---|------------------|
| • (UG) Intelligent Networking and Computing | Spring 2023 |
| • (UG) Introduction to Wireless Sensor Networks | Spring 2017-2022 |
| • (UG) Internet Programming (in JAVA) | Fall 2017-2023 |
| • (PG) Guidance on Academic Writing | Fall 2021-2023 |

STUDENT SUPERVISION

Ph.D. Students @ The Hong Kong University of Science and Technology (Guangzhou):

- **Xiaowen Xu** (2024.09 -)
B.E. of Computer Science and Technology, Shenzhen University, Shenzhen, China, 2023 (“[Excellent Undergraduate of Shenzhen University](#)”)
- **Jiyu Wang** (2025.02 -)
M.E. of Transportation Engineering, Sun Yat-sen University (Shenzhen campus), Shenzhen, China, 2024
- **Zhenyu Yang** (2026.02 -)
M.E. of Computer Science and Technology, Shenzhen University, Shenzhen, China, 2025

Research Assistant @ The Hong Kong University of Science and Technology (Guangzhou):

- **Jinsong Luo** (2025.07 -)
B.E. of Artificial Intelligence, Central South University, Changsha, China, 2025

- **Fansheng Sun** (2025.10 -)
B.E. of Computer Science and Technology, Xi'an Jiaotong University, Xi'an, China, 2026
- **Yuhang Guan** (2025.08 -)
B.E. of Computer Science and Technology, Shandong University, Qingdao, China, 2025

Alumni of The Hong Kong University of Science and Technology (Guangzhou):

1. **Zitong Yu**: Research Assistant (2025.08 - 2026.07); UG @ China University of Geosciences (Wuhan) → [MSc @ University of California, Berkeley](#)
2. **Guangyu Wang**: Research Assistant (2025.08 - 2025.11); UG @ Dongbei University of Finance and Economics → [Ph.D. student @ HKUST \(GZ\)](#)
3. **Yanlin YU**: Research Assistant (2025.08 - 2025.11); MSc @ University of Warwick → [Ph.D. student @ HKUST \(GZ\)](#)

Alumni of Shenzhen University:

1. **Zengyang Gong** (Master 2017.09 - 2020.06)
 - [Award: "Excellent Graduate of Shenzhen University"](#)
 - Publications: IEEE ICDE'20, IEEE ICCCN'20, IEEE IOTJ'22
 - Employment: Ph.D candidate in The Hong Kong university of Science and Technology (HKUST)
 - B.S. of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan, China, 2017
2. **Mingliang Huang** (Master 2017.09 - 2020.06)
 - Publications: IEEE MSN'19
 - Employment: Sangfor Technologies
 - B.S. of Computer Science and Technology, Lanzhou University, Lanzhou, China, 2017
3. **Jiangzhou Li** (Master 2018.09 - 2021.06)
 - Publications: IEEE TITS'22
 - Employment: Sangfor Technologies
 - B.S. of Software Engineering, Qingdao University, Qingdao, China, 2018
4. **Baofu Huang** (Master 2018.09 - 2021.06)
 - [Award: "Best Paper Award" of IEEE ICPADS'20](#)
 - Publications: IEEE HPCC'20, IEEE ICPADS'20
 - Employment: Tencent
 - B.E. of Computer Science, Guangdong University of Finances and Economics, Guangzhou, China, 2018
5. **Weixin Lin** (Master 2019.09 - 2022.06)
 - Publications: CJC'22
 - Employment: JinTeng Technologies
 - B.E. of Computer Science and Technology, Shenzhen University, Shenzhen, China, 2019
6. **Junhong Zheng** (Master 2019.09 - 2022.06)
 - Publications: DASFAA'21, IEEE TMC'23
 - Employment: Tencent
 - B.S. of Software Engineering, Zhuhai College of Science and Technology, Zhuhai, China, 2019
7. **Jiancong Liu** (B.S. 2018.09 - 2022.06)
 - [Award: "Excellent Undergraduate of Shenzhen University"](#)
 - Publications: IEEE TMC'24
 - Employment: Master student in Harbin Institute of Technology (Shenzhen)
 - B.E. of Computer Science and Technology, Shenzhen University, Shenzhen, China, 2022
8. **Hongquan Zhang** (Master 2020.09 - 2023.06)
 - [Award: "Excellent Graduate of Shenzhen University"](#)
 - Publications: IEEE TMC'24
 - Employment: HUAWEI
 - B.E. of Internet of Things Engineering, Hunan Agricultural University, Changsha, China, 2020
9. **Meiquan Lai** (Master 2020.09 - 2023.06)
 - Publications: IEEE ICPADS'24
 - Employment: SICS
 - B.S. of Mathematics, Shenzhen University, Shenzhen, China, 2020

10. **Yingqian Zhou** (Master 2020.09 - 2023.06)
 - Publications: IEEE TKDE'25
 - Employment: BOCOM
 - B.E. of Computer Science and Technology, Guangdong University of Technology, Guangzhou, China, 2020
11. **Jiayue Ruan** (Master 2020.09 - 2023.06)
 - Employment: BYD company
 - B.E. of Computer Science and Technology, North China University of Technology, Beijing, China, 2020
12. **Wenxuan Xia** (B.S. 2020.09 - 2023.06)
 - Award: [“Excellent Undergraduate of Shenzhen University”](#)
 - Employment: Ph.D. student at HKUST (GZ))
 - B.S. of Software Engineering, Shenzhen University, Shenzhen, China, 2023
13. **Yaqi Huang** (Master 2021.09 - 2024.06)
 - Publications: GPC'25
 - Employment: Tencent Music
 - B.E. of Internet of Things Engineering, Hunan Agricultural University, Changsha, China, 2021
14. **Guofeng Ouyang** (Master 2021.09 - 2024.06)
 - Publications: IEEE TMC'25
 - Employment: FUTU
 - B.E. of Electronical Information Science and Technology, Zhaoqing University, Zhaoqing, China, 2021
15. **Jinye Lin** (Master 2021.09 - 2024.06)
 - Publications: IEEE ICDM'24
 - Employment: FUTU
 - B.E. of Internet of Things Engineering, Shenzhen University, Shenzhen, China, 2021
16. **Xiaosi Liu** (Master 2022.09 - 2025.06)
 - Publications: IEEE TMC'25, IEEE TKDE'25
 - Employment: BYD company
 - B.E. of Software Engineering, Nanchang University, Nanchang, China, 2022
17. **Zhiyu Xia** (Master 2022.09 - 2025.06)
 - Employment: HSBC Group
 - B.E. of Computer Science and Technology, Shantou University, Shantou, China, 2022
18. **Zhengze Sun** (Master 2022.09 - 2025.06)
 - Publications: IEEE TITS'25, GPC'25
 - Employment: HUAWEI
 - B.E. of Data Science and Big Data Technology, Wuhan University of Technology, Wuhan, China, 2022
19. **Zhenyu Yang** (Master 2022.09 - 2025.06)
 - Publications: IEEE TMC'26
 - Employment: PhD @ HKUST(GZ)
 - B.E. of Computer Science and Technology, Henan University, Kaifeng, China, 2022
20. **Haotian Zhong** (B.E 2022.06 - 2025.06)
 - Award: [“Excellent Undergraduate of Shenzhen University”](#)
 - Employment: Ph.D. student at Zhejiang University
 - B.E. of Software Engineering, Shenzhen University, Shenzhen, China, 2025
21. **Bolin Zhang** (Master 2023.09 - 2026.06)
 - Award: [“Excellent Undergraduate of Shenzhen University”](#)
 - Publications: ACM CIKM'25
 - Employment: Meituan
 - B.E. of Software Engineering, Shenzhen University, Shenzhen, China, 2023
22. **Chang Liu** (Master 2023.09 - 2026.06)
 - Employment: Shopee
 - B.E. of Internet of Things Engineering, Jinan University, Guangzhou, China, 2023

23. **Yuhao Ouyang** (Master 2023.09 - 2026.06)

- Publications: IEEE TMC (under revision)
- Employment: ByteDance
- B.E. of Data Science and Big Data Technology, Beijing Forestry University, Beijing, China, 2023

ACADEMIC COMMUNITY SERVICE

Editorial Board:

- Associate Editor: **IEEE Transactions on Emerging Topics in Computational Intelligence**, 2026.07 - Now
- Young Editorial Board Member: **Journal of Social Computing**, 2024.02 - Now
- Lead Guest Editor: *Special Issue on Ubiquitous Sensing and Computing for Intelligent Transportation* in **Data Science for Transportation**

Membership:

- **IEEE Senior Member** (since 2024)
- **CCF Senior Member** (since 2023)
- Member of ACM (since 2018)
- Member of ACM SIGMOBILE (since 2021)
- Executive member of CCF Ubiquitous Computing Technical Committee (since 2022)
- Executive member of CCF Internet of Things Technical Committee (since 2023)
- Executive member of CCF Intelligent Transportation Division (since 2023)
- Executive member of ACM SIGSpatial China (since 2023)

Conference Organization Chair:

- Special Session Chair: IEEE CSCWD 2022
- Local Chair: EAI ICECI 2021
- Conference Finance Chair: IEEE ICPADS 2017
- Conference Finance Chair: IEEE ICPADS 2018

Technical Program Committee Member:

- 2027: ACM KDD, AAAI (**Senior PC**)
- 2026: ACM KDD, IEEE ICDE (demo track), VLDB (demo track), The 20th International Conference on Green, Pervasive, and Cloud Computing (GPC), The 18th International Conference on Networking, Architecture, and Storage (NAS), IEEE ICPADS, IEEE MSN
- 2025: ACM CSCW (**Associate Chair**), ACM KDD (**Outstanding Reviewer** for August Cycle), IEEE MSN, Australasian Database Conference (ADC)
- 2024: ACM KDD, IEEE ICDE (demo track), IEEE MSN
- 2023: ACM CIKM, IEEE ICPADS
- 2022: ACM KDD, ACM WSDM, IEEE ICPADS
- 2021: IEEE ICDCS, IEEE MASS
- 2020: IEEE ICDCS, IEEE HiPC
- 2019: IEEE MASS, IEEE MSN
- 2018: IEEE ICPADS, IEEE ICCCN
- 2017: IEEE ICPADS

Transaction and Journal Reviewer (selected):

- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- IEEE Transactions on Mobile Computing (TMC)
- IEEE Transactions on Intelligent Transportation Systems (TITS)
- IEEE/ACM Transactions on Networking (TON)
- ACM Transactions on Spatial Algorithms and Systems (TSAS)
- IEEE Transactions on Wireless Communications (TWC)
- IEEE Transactions on Parallel and Distributed Systems (TPDS)
- ACM Transactions on Sensor Networks (TOSN)
- IEEE Transactions on Communications (TOC)
- IEEE Internet of Things Journal (JIOT)
- IEEE Communications Magazine
- IEEE Computational Intelligence Magazine
- CCF Transactions on Pervasive Computing and Interaction (TPCI)
- Elsevier Pervasive and Mobile Computing (PMC)
- Ad Hoc & Sensor Wireless Networks (AHSWN)
- Elsevier Computer Networks